SECTION A

ATTEMPT ANY THREE QUESTIONS

1.(a) A compound **N** contains **92.31%** carbon and **7.69%** hydrogen. Determine the **empirical formula of N**. (02 marks)

a) Compound **N** burns with a **sooty flame** and has vapour density **39**.Determine the **molecular formula of compound N** and write it's **molecular structure**. (03 marks)

b) Write **equation** and **indicate a mechanism** for the reaction between **compound N** and:

i. Concentrated nitric and sulphuric acid. (05 marks)

ii. Ethanoyl chloride. (04 marks)

iii. Propene. (04 marks)

c) Write equation of reaction to show how **compound N** can be synthesized from cyclohexane. (02 marks)

2.(a) Define the following terms: (@02 marks)

i. **Heat of reaction.**

ii. **Enthalpy of formation.**

iii. **Hess's law.**

iv. **Heat of combustion**.

(b) (i) Assuming that kerosene is $C_{11}H_{24}$, Calculate the **standard enthalpy change** when one mole of kerosene burns completely in oxygen, using tha data below: (06 marks)

$C_{11}H_{24} = {}^{-}327KJ/mol$, $CO_2 = {}^{-}394KJ/mol$, $H_2O = {}^{-}286KJ/mol$

ii. Comment on your answer in b(i). (01 mark)

iii. State three factors affecting enthalpy of reaction. (03 marks)

3.(a) What is meant by the term **diagonal relationship**? (02 marks)

(b) Explain why beryllium shows **different properties** from the **rest of group** (II) elements.

(c) State **three properties** in which:

i. **Beryllium** differs from the **rest of group (II) elements**. (03 marks)

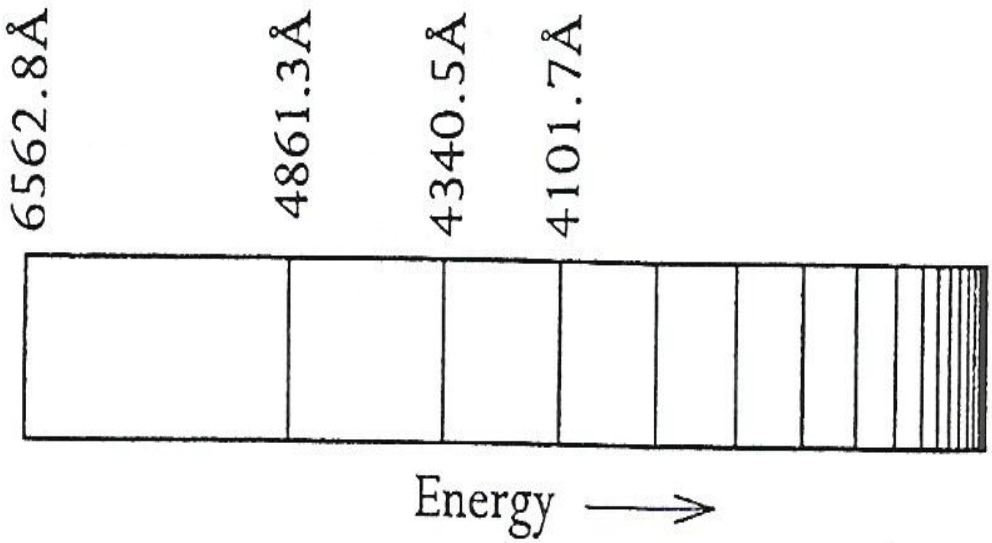
ii. **Beryllium** resembles **aluminium**. (03 marks)

(d) State and explain how the **solubility** of the following compounds of group (II) elements **vary down** the group.

i. **Hydroxide** (03½ marks)

ii. **Sulphate** (03½ marks)

4. (a) (i) What is spectrum of an element? (02 marks)
   (ii) Explain how hydrogen spectrum is formed. (05 marks)
   (b) The diagram below shows the atom spectrum of an element.



   (i) Explain why the emission lines get closer. (03 marks)
   (ii) When an electron returns from energy level **X** to energy level **2**, green light was emitted. Determine energy level **X**. (03 marks)
   [Rydberg's constant, R = 109678cm^{-1}, 1cm = 10^{7}nm]
   (iii) State three applications of emission spectrum. (03 marks)
   (c) Borh theory has always been used to explain hydrogen spectrum. Identify the: (@02 marks)
   i. Advantages of the theory.
   ii. Disadvantages of the theory.

SECTION B

ATTEMPT ONLY TWO QUESTIONS IN THIS SECTION.

5. Discuss the following reactions:
   a) Methylbenzene with chlorine.
   b) Ethanol with sulphuric acid.
   c) Chloromethane with potassium hydroxide.

   You answer should include
   i. Equations for the reactions.
   ii. Acceptable mechanisms for the reactions. (20 marks)

6. (a) Define the terms: (@01 mark)
   i. **Relative atomic mass.**
   ii. **Isotopes.**

(b) A naturally occurring element Q has two isotopes, $_{20}Q$ and $_{22}Q$. The relative atomic mass of Q is **20.2**.
   i. Calculate the **relative abundances** of the different isotopes of Q. (03 marks)
   ii. Briefly describe how the **mass spectrometer** can be used to obtain the relative atomic mass of Q. [***No diagram required***] (06 marks)

(c) State
   (i) **Two** uses of isotopes. (02 marks)
   (ii) **One** limitation in using mass spectrometer for relative atomic mass determination. (01 mark)

(d) One of the reasons why isotopes of elements undergo radioactive decay is to achieve stability in the nuclei. The graph in figure 1 shows a plot of number of neutrons against number of protons for stable nuclei.

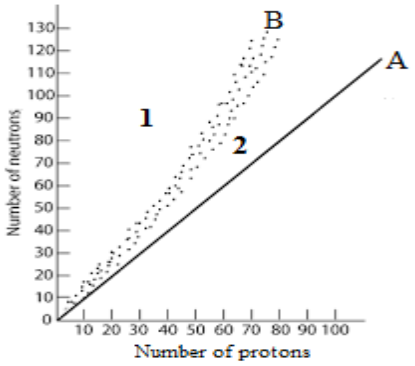


Figure 1

State:
   i. Two factors that determine the stability of a nucleus of an atom. (01 mark)
   ii. What **line A** represents. (0½ mark)
   iii. What **region B** represents. (0½ mark)

(e) Briefly describe how **isotopes in region 1** and **2** in figure 1 can achieve stability. (04 marks)

7. **Mg, Al, Si, S,** & **Cl** belong to **period 3** of the periodic table.
   a) Describe the **reaction** of the **period 3** elements with: (@08½ marks)
      i. Water ii. Sodium hydroxide solution.
   b) Explain why an aqueous solution of aluminium chloride is **acidic**. (03 marks)

8.(a) State what is meant by the following terms: (@01 mark)

i. Radioactivity.

ii. Law of alpha decay.

(b) The table below shows data for the radioactive decay of element Z.

Time (Hrs)	0.0	5.0	10.0	15.0	20.0	25.0	30.0
Activity (Count/min)	25.00	23.0	21.25	19.50	18.00	16.50	15.25

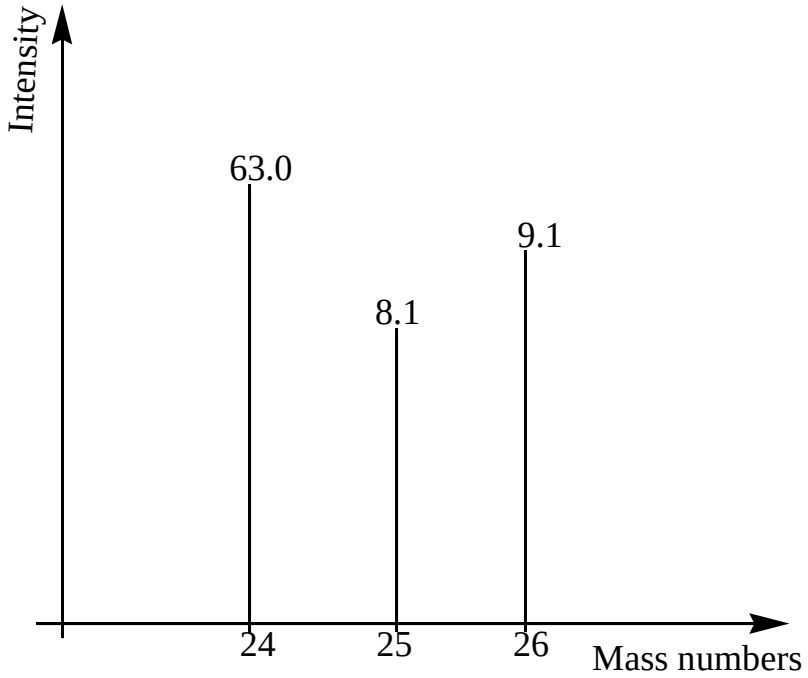
i. Plot a graph of activity against time. (03 marks)

ii. Determine the value of half-life of element Z. (01½ marks)

iii. Determine the decay constant and state its unit. (02 marks)

(c)(i) Distinguish between isotopy and allotropy. (02 marks)

(ii)The figure below shows the mass spectrum of magnesium. The heights of the three peaks and the mass numbers of the isotopes are shown in the figure below:



Calculate the relative atomic mass of magnesium. (05 marks)

(d)Account for the structure of atom using Rutherford concepts. (04½ marks)

THE PERIODIC TABLE

1	2											3	4	5	6	7	8
1.0 H 1																1.0 H 1	4.0 He 2
6.9 Li 3	9.0 Be 4											10.8 B 5	12.0 C 6	14.0 N 7	16.0 O 8	19.0 F 9	20.2 Ne 10
23.0 Na 11	24.3 Mg 12											27.0 Al 13	28.1 Si 14	31.0 P 15	32.1 S 16	35.4 Cl 17	40.0 Ar 18
39.1 K 19	40.1 Ca 20	45.0 Sc 21	47.9 Ti 22	50.9 V 23	52.0 Cr 24	54.9 Mn 25	55.8 Fe 26	58.9 Co 27	58.7 Ni 28	63.5 Cu 29	65.7 Zn 30	69.7 Ga 31	72.6 Ge 32	74.9 As 33	79.0 Se 34	79.9 Br 35	83.8 Kr 36
85.5 Rb 37	87.6 Sr 38	88.9 Y 39	91.2 Zr 40	92.9 Nb 41	95.9 Mo 42	98.9 Tc 43	101 Ru 44	103 Rh 45	106 Pd 46	108 Ag 47	112 Cd 48	115 In 49	119 Sn 50	122 Sb 51	128 Te 52	127 I 53	131 Xe 54
133 Cs 55	137 Ba 56	139 La 57	178 Hf 72	181 Ta 73	184 W 74	186 Re 75	190 Os 76	192 Ir 77	195 Pt 78	197 Au 79	201 Hg 80	204 TI 81	207 Pb 82	209 Bi 83	209 Po 84	210 At 85	222 Rn 86
223 Fr 87	226 Ra 88	227 Ac 89															

139 La 57	140 Ce 58	141 Pr 59	144 Nd 60	147 Pm 61	150 Sm 62	152 Eu 63	157 Gd 64	159 Tb 65	162 Dy 66	165 Ho 67	167 Er 68	169 Tm 69	173 Yb 70	175 Lu 71
227 Ac 89	232 Th 90	231 Pa 91	238 U 92	237 Np 93	244 Pu 94	243 Am 95	247 Cm 96	247 Bk 97	251 Cf 98	254 Es 99	257 Fm 100	256 Md 101	254 No 102	260 Lw 103

♥ ===END===

WELCOME TO SENIOR FIVE, YEAR 2019

This is the last page of the printed paper, Page 05